

Emmanuel Tega Agbragu

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SUMMARY

Security Engineer with 4+ years of experience deploying and supporting security infrastructure. Hands-on **Wazuh** experience covering SCA, vulnerability detection, alert analysis, remediation, and NIST compliance validation. Proficient in Linux, TCP/IP, firewalls, SIEM, IDS/IPS, incident response, and automation, with exposure to Git, Ansible, Docker, Kubernetes, cloud platforms, virtualization, and Elastic Stack.

EXPERIENCE

Security Researcher

Jan 2025 – Apr 2026

The Upanzi Network, Carnegie Mellon University | Kigali, Rwanda

- Conducted vulnerability assessments across **5** research platforms, producing technical documentation and proof-of-concept evidence to support remediation of critical network and application issues.
- Developed **Python and Bash automation scripts** for endpoint enumeration, vulnerability detection, and large-scale asset analysis.
- Tuned security monitoring configurations to reduce false positives and verified remediation through collaborative retesting with technical teams.
- Identified a **Critical (CVSS 9.1)** BOLA/IDOR vulnerability chain and emerging prompt-injection risks, translating findings into actionable security hardening guidance.

Graduate Teaching Assistant

Sep 2024 – Jan 2025

Carnegie Mellon University Africa | Kigali, Rwanda

- Delivered labs and technical presentations in Ethical Hacking and Cybersecurity Operations, reinforcing practical skills in network security, **TCP/IP**, incident response, and security operations.

Cybersecurity Engineer

Apr 2021 – Jun 2023

Lumenave International Limited | Lagos, Nigeria

- Deployed and supported IDS/IPS, SIEM, and EDR solutions for banking clients, reducing malware infections by **40%** through Linux system hardening and detection-policy tuning.
- Configured **firewall rules**, Symantec DLP policies, and secure network controls to protect sensitive data and maintain reliable internet connectivity.
- Implemented CyberArk PAM and least-privilege access controls aligned with ISO 27001 and SOC 2 requirements.
- Led a team of **5 engineers** delivering security projects and managed security services, maintaining **100% operational efficiency** through automation and orchestration.

SELECTED PROJECTS

Wazuh Security Operations Project — Cybersecurity Operations (04-722)

2024

Wazuh, SIEM, SCA, PowerShell | Carnegie Mellon University Africa

- Operated a **Wazuh deployment** monitoring a Windows 11 VM; triaged Security Configuration Assessment and vulnerability alerts, including critical CVEs, and prioritized remediation by severity and impact.
- Automated CIS-aligned configuration remediation with PowerShell and used Wazuh compliance mapping to validate **NIST 800-53** controls AC-7, AU-5, and SC-8.

Securing the Digital State — Research Report

2025

Python, HTTP, Automation | Upanzi Network

- Assessed the security posture of **21,782** government sub-domains across 54 African countries; led HTTP-method enumeration and built Python automation for mass asset discovery and scanning.

TECHNICAL SKILLS

Security Operations	Wazuh, SIEM, SCA, Vulnerability Detection, Alert Triage, IDS/IPS, EDR, Firewall, VPN, Elastic Stack
Linux & Networking	Linux Hardening, TCP/IP, HTTP, DNS, Routing, Switching, VLAN, NAT, SD-WAN, Network Troubleshooting
Programming & Automation	Python, Bash, PowerShell, Git, Ansible, Terraform, CloudFormation, CI/CD
Cloud & Infrastructure	AWS, Azure, GCP, Docker, Kubernetes, Virtualization
Frameworks & Compliance	OWASP Top 10, MITRE ATT&CK, NIST 800-53, ISO 27001, SOC 2, PCI-DSS, CIS Benchmarks
Certifications	CEH Master, eJPT, AWS Certified Cloud Practitioner, OSCP (In Progress)

EDUCATION

MSc Information Technology

Aug 2023 – May 2025

Carnegie Mellon University | GPA: 3.80/4.0 | Mastercard Foundation Partial Scholarship

BSc Pure and Applied Physics — First Class Honours

Mar 2014 – May 2018

University of Benin | Nigeria | GPA: 4.54/5.0 | Top 1% of graduating class

CERTIFICATIONS AND ACHIEVEMENTS

CEH Master • eJPT • AWS Certified Cloud Practitioner • OSCP (In Progress) • Top 10 Finalist, Greenfist CTF 2023 • Mastercard Foundation Scholarship 2023 • First Class Honours, Top 1%, 2018